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Full publication list

Peer-reviewed journals

1. 2026, Rubin Observatory Science Pipelines Developers, Y. AlSayyad, R. Armstrong [and 83 others, including **N. Caplar**]
   The LSST Science Pipelines Software: Optical Survey Pipeline Reduction and Analysis Environment, submitted
   arXiv:2610.22613

2. 2026, A. Ghosh, D. Oldag, M. Tauraso [and 29 others, including **N. Caplar**]
   Hyrax: An Extensible Framework for Rapid ML Experimentation and Unsupervised Discovery in the Era of Rubin, Roman, and Euclid, submitted to AJ
   arXiv:2605.18959

3. 2026, C. O. Chandler, P. H. Bernardinelli, M. Juric [and 303 others, including **N. Caplar**]
   NSF-DOE Vera C. Rubin Observatory Observations of Interstellar Comet 3I/ATLAS (C/2025 N1), ApJL, 1001, L35
   10.3847/2041-8213/ae4b3a

4. 2026, Vera C Rubin Observatory Team, [and 327 others, including **N. Caplar**]
   The Vera C. Rubin Observatory Data Preview 1, AJ, 171, 360
   10.3847/1538-3881/ae521f

5. 2026, A. B. Kovačević, D. Ilić, U. Meštrić [and 31 others, including **N. Caplar**]
   QhX Catalog of Coherent and Quasi-coherent Active Galactic Nucleus Variability from Vera C. Rubin LSST DP1, ApJS, 285, 61
   10.3847/1538-4365/ae8444

6. 2026, Y. D. Wang, E. C. Bellm, R. I. Hynes [and 9 others, including **N. Caplar**]
   Optical Counterparts to X-Ray Sources in LSST DP1, ApJS, 284, 13
   10.3847/1538-4365/ae5247

7. 2026, L. Palaversa, E. Donev, Z. Ivezic, K. Mrakovcic, **N. Caplar**, [and 94 others]
   photoD with Rubin's Data Preview 1: First stellar photometric distances and faint blue star deficits: Stellar distances with Rubin's DP1, A&A, 707, A333
   10.1051/0004-6361/202557219

8. 2025, K. Malanchev, M. DeLucchi, **N. Caplar**, [and 56 others]
   Variability-finding in Rubin Data Preview 1 with LSDB, submitted to AJ
   arXiv.2506.23955

9. 2025, A. Tzanidakis, J. R. A. Davenport, **N. Caplar**, [and 8 others]
A Systematic Search for Main-sequence Dipper Stars Using the Zwicky Transient Facility, ApJ, 991, 118
10.3847/1538-4357/adf5bb

10. 2025, L Palaversa, Z. Ivezic, **N. Caplar**, K. Mrakovcic, B. Abel, O. Razim, F. Matkovic, C. Yablonski, T. Saric, T. Jurkic, S. Campos, M. DeLucchi, D. Jones, K. Malanchev, A. Malz, S. McGuire, M. Juric
PhotoD with LSST: Stellar Photometric Distances Out to the Edge of the Galaxy, AJ
doi: 10.3847/1538-3881/ada3c2

11. 2024, K. G. Iyer, J. S. Speagle, **N. Caplar**, J. C. Forbes, E. Gawiser, J. Leja, S. Tacchella
Stochastic Modelling of Star Formation Histories III. Constraints from Physically-Motivated Gaussian Processes, ApJ
doi:10.3847/1538-4357/acff64

12. 2023, C. Burke, Y. Shen, X. Liu, P. Natarajan, **N. Caplar**, J. Bellovary, Z. Wang
Dwarf AGNs from Variability for the Origins of Seeds (DAVOS): Intermediate-mass black hole demographics from optical synoptic surveys, MNRAS, 518, 1880
doi: 10.1093/mnras/stac2478

13. 2022, A. B. Kovacevic, V. Radovic, D. Ilic, [and 22 others, including **N. Caplar**]
The LSST era of supermassive black holes accretion-disk reverberation mapping, APJS
doi:10.3847/1538-4365/ac88ce

14. 2021, A. Kovacevic, D. Ilic, L. Popovic, V. Radovic, I. Jankov, I. Yoon, **N. Caplar**, I. Cvorovic-Hajdinjak, S. Smic
On possible proxies of AGN light-curves cadence selection in future time domain surveys, MNRAS
doi.org/10.1093/mnras/stab1595

15. 2020, K. G. Iyer, S. Tacchella, S. Genel, C. C. Hayward, L. Hernquist, A. M. Brooks, **N. Caplar**, R. Dave, B. Diemer, J. C. Forbes, E. Gawsier, R. S. Somerville, T. K. Starkenburg
The Diversity and Variability of Star Formation Histories in Models of Galaxy Evolution, MNRAS
doi.org/10.1093/mnras/staa2150

16. 2020, S. Tacchella, J. C. Forbes, **N. Caplar**
Stochastic modelling of star-formation histories II: star-formation variability from molecular clouds and gas inflow, MNRAS, 497, 698T
doi.org/10.1093/mnras/staa1838

17. 2020, I. Delvecchio, E. Daddi, J. Mullaney, E. Bernhard, L. Grimmett, R. Carraro, A. Cimatti, G. Zamorani, **N. Caplar**, D. Elbaz, G. Rodighiero
The evolving AGN duty cycle in galaxies since z $\sim$ 3 as encoded in the X-ray luminosity function, ApJ, 892, 17D
doi.org/10.3847/1538-4357/ab789c

18. 2020, **N. Caplar**, T. Penna, S. Johnson, J. Greene
Nonstationarity of AGN variability: the only way to go is down!, ApJL, 2020, 889L, 29C
doi.org/10.3847/2041-8213/ab6a11

19. 2019, *(corresponding author)* L. Sartori, B. Trakhtenbrot, K. Schawinski, **N. Caplar**, E. Treister, C. Zhang
A forward modelling approach to AGN variability – method description and early applications, ApJ, 883, 139S
doi.org/10.3847/1538-4357/ab3c55

20. 2019, **N. Caplar**, S. Tacchella
Stochastic modeling of star-formation histories I: the scatter of the star-forming main sequence, MNRAS, 487, 3845C
doi.org/10.1093/mnras/stz1449

21. 2018, L. Sartori, K. Schawinski, B. Trakhtenbrot, **N. Caplar**, E. Treister, M. Koss, M. Urry, C. Zhang
A model for AGN variability on multiple time-scales, MNRAS, 476L, 34S
doi.org/10.1093/mnrasl/sly025

22. 2018, **N. Caplar**, S. Lilly, B. Trakhtenbrot
AGN evolution from galaxy evolution viewpoint - II, ApJ, 867, 148C
doi.org/10.3847/1538-4357/aae691

23. 2017, **N. Caplar**, S. J. Lilly, B. Trakhtenbrot
Optical variability of AGN in the PTF/iPTF survey, ApJ, 834, 111C
doi.org/10.3847/1538-4357/834/2/111

24. 2017, A. Weigel, K. Schawinski, **N. Caplar**, A. Carpineti, R. Hart, S. Kaviraj, W. Keel, S. Kruk, C. Lintott, R. Nichol, B. Simmons, R. Smethurst
Galaxy Zoo: Major galaxy mergers are not a significant quenching pathway, APJ, 845, 145W
doi.org/10.3847/1538-4357/aa8097

25. 2017, A. Weigel, K. Schawinski, **N. Caplar**, O. I. Wong, T. Ezequiel, B. Trakhtenbrot
AGN and their host galaxies in the local Universe: Two mass-independent Eddington ratio distribution functions characterize black hole growth, ApJ, 845, 134W
doi.org/10.3847/1538-4357/aa803b

26. 2017, **N. Caplar**, S. Tacchella, S. Birrer
Quantitative evaluation of gender bias in astronomy, NatAs, 1, 0141
doi.org/10.1038/s41550-017-0141

27. 2015, **N. Caplar**, S. J. Lilly, B. Trakhtenbrot
AGN evolution from a galaxy evolution viewpoint, ApJ, 811, 148C
doi.org/10.1088/0004-637X/811/2/148

28. 2013, **N. Caplar**, H. Stefancic
Generalized models of unification of dark matter and dark energy, Phys. Rev. D, 87, 023510
doi.org/10.1103/PhysRevD.87.023510

Research Notes of the AAS

1. 2025, T. M. Wainer, J. R. A. Davenport, E. C. Bellm, Y. Wang, **N. Caplar**, [and 12 others]
   Crowded Field Photometry with Rubin: Exploring 47 Tucanae with Data Preview 1, RNAAS, 9, 171
   10.3847/2515-5172/adecef

2. 2025, J. L. Carlin, P. S. Ferguson, A. K. Vivas, **N. Caplar**, [and 1 other]
   An Outer-disk SX Phe Variable Star in Rubin Data Preview 1, RNAAS, 9, 161
   10.3847/2515-5172/adeab0

Conference proceedings

1. 2024, **N. Caplar**, W. Beebe, D. Branton, S. Campos, A. Connolly, M. DeLucchi, D. Jones, M. Juric, J. Kubica, K. Malanchev, R. Mandelbaum, S. McGuire
   Using LSDB to enable large-scale catalog distribution, cross-matching, and analytics
   Proceedings of XXXIV Astronomical Data Analysis Software & Systems (ADASS) conference, November 10-14 2024, Valletta, Malta; 10.48550/arXiv.2501.02103
2. 2022, **N. Caplar**, R. Lupton, J. E. Gunn, H. Siddiqui, P. Price, C. Loomis, A. L. Fur, J. E. Meyers
   Prime focus spectrograph (PFS) for the Subaru Telescope: 2D modeling of the point spread function
   Proc. SPIE 12184, Ground-based and Airborne Instrumentation for Astronomy IX, 1218470 (29 August 2022); doi.org/10.1117/12.2629364
3. 2022, Shian-Yu Wang, Masahiko Kimura, Chi-Huang Yan , [and 29 others, including **N. Caplar**]
   Prime focus spectrograph (PFS) for the Subaru Telescope: the prime focus instrument
   Proc. SPIE 12184, Ground-based and Airborne Instrumentation for Astronomy IX, 121846R (29 August 2022); doi.org/10.1117/12.2629098
4. 2022, K. Breivik, A. J. Connolly, K. E. S Ford, [and 94 others, including **N. Caplar**]
   From Data to Software to Science with the Rubin Observatory LSST
   https://arxiv.org/abs/2208.02781
5. 2018, T. Naoyuki , T. Naruhisa, A. Shimono, [and 111 others, including **N. Caplar**]
   Prime Focus Spectrograph (PFS) for the Subaru telescope: ongoing integration and future plans, Proceedings of the SPIE, Volume 10702, id. 107021C 12 pp.
6. 2013, **N. Caplar**, M. Suznjevic, M. Matijasevic
   Analysis of players' in-game performance vs rating: Case study of Heroes of Newerth, Foundation of Digital games 2013, pp. 237-244